EE561 Advanced Static Power Conversion

Multilevel Converters: Building the Levels — Quiz

about 8 minutes · closed book · calculator allowed · prepared by Sefer İdacı

Name: Student ID: Date:

Answer both questions in the space provided; show your reasoning. Each question is worth 25 points (total 50). Both questions cover the topologies talk: device ratings, NPC and T-type legs.

Question 1 · Device selection

NUMERICAL DESIGN **25 pts**

A PV string inverter has a DC link of **1100 V**. In steady state a switch may block at most **2/3 of its voltage rating**; the available classes are 650 V, 1200 V, 1700 V and 3300 V.

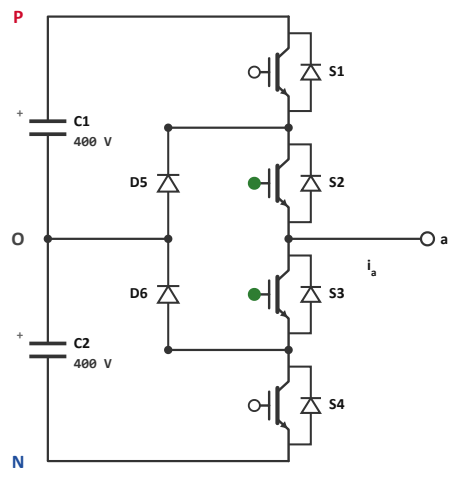
a. Complete the table. [15]

Leg / device	Voltage blocked by the device [V]	Required class [V]
2-level leg, each switch		
3-level NPC leg, each switch		
3-level T-type leg, outer switches S1, S4		
3-level T-type leg, inner switches S2, S3		

b. Give one advantage and one disadvantage of the T-type leg compared with the NPC leg at this DC link. [10]

Question 2 · NPC leg in the zero state

CONCEPT AND CRITIQUE **25 pts**



3-level NPC leg, V_{dc} = 800 V, state O (gated switches marked green).

The leg is in state **O** and the phase current is negative, $i_a < 0$ (flowing *into* the leg).

a. Draw the current path on the figure and name the conducting devices. [7]

b. Does the leg draw current from the midpoint O or push current into it? [5]

c. Which devices block voltage, and how much? [6]

d. A designer proposes to switch directly from P to N "to save one commutation". Critique this proposal. [7]